

LANGDON BAY, United Kingdom

WMO: 037960

Lat: 51.134N

Lon: 1.343E

Elev: 384

StdP: 14.49

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.8	30.7	22.3	17.0	31.9	24.8	19.1	33.1	41.7	42.4	36.7	44.5	17.0	330	0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	75.0	65.5	71.6	63.5	68.8	62.3	66.6	72.6	64.9	69.5	63.5	67.3	10.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.7	69.1	63.0	87.3	66.8	61.9	84.0	65.5	31.4	72.9	30.1	69.9	29.1	67.3	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.2	30.6	27.6	DB	25.6	82.0	2.7	4.6	23.7	85.3	22.1	87.9	20.6	90.5	18.7	93.8	(4)
(5)			WB	24.2	70.1	2.7	1.9	22.2	71.5	20.6	72.6	19.1	73.7	17.1	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	41.1	40.9	43.9	48.0	53.3	57.6	61.9	62.4	58.9	53.9	46.8	42.5
	DBStd	8.98	5.02	5.04	4.58	4.82	4.61	4.08	4.03	3.64	3.65	4.41	5.14	5.55
	HDD50	1207	276	254	194	91	18	0	0	0	0	16	120	238
	HDD65	5162	741	674	655	510	364	226	115	101	185	346	547	698
	CDD50	1566	0	0	4	32	119	228	369	383	268	136	23	4
	CDD65	46	0	0	0	0	0	5	19	19	3	0	0	0
	CDH74	146	0	0	0	0	2	18	63	58	5	0	0	0
	CDH80	17	0	0	0	0	0	2	9	6	0	0	0	0
Wind	WSAvg	14.4	16.9	16.1	15.1	13.7	13.6	12.4	11.6	12.1	13.3	15.0	16.0	16.7
Precipitation	PrecAvg	27.8	2.6	2.0	1.6	1.8	1.9	2.0	2.0	2.2	2.2	3.3	3.4	2.8
	PrecMax	38.6	5.6	4.7	4.1	5.3	4.9	4.6	3.8	4.9	4.1	8.3	6.4	4.8
	PrecMin	21.3	0.7	0.4	0.2	0.2	0.3	0.6	0.6	0.5	0.2	0.5	1.2	0.6
	PrecStd	4.1	1.1	1.1	0.9	1.1	1.1	1.2	0.9	1.3	1.1	1.7	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	52.9	59.1	68.2	71.4	76.4	79.9	80.0	73.3	66.3	59.2	54.6
		MCWB	50.1	48.3	52.1	56.7	61.6	65.8	67.9	66.8	64.8	60.9	56.2	52.8
	2%	DB	50.7	50.3	54.9	61.2	67.3	70.4	74.9	74.8	68.4	63.2	57.0	53.1
		MCWB	49.1	47.4	49.1	53.2	59.1	62.4	65.7	65.6	62.5	59.5	54.8	51.9
	5%	DB	49.5	49.0	51.9	57.6	63.6	66.9	71.5	70.7	66.3	61.5	55.6	52.0
		MCWB	48.2	46.8	47.7	51.5	57.0	60.6	63.6	63.8	61.2	58.6	53.5	50.6
	10%	DB	48.3	47.8	49.9	54.6	60.4	64.2	68.4	68.0	64.5	60.1	54.0	50.6
		MCWB	46.9	46.0	46.8	49.9	55.3	59.2	62.4	62.2	60.0	57.5	51.9	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	49.9	52.7	57.9	63.3	66.9	69.6	69.1	65.8	62.0	57.3	53.7
		MCDB	51.6	50.9	57.5	65.3	69.4	74.5	77.5	76.9	70.2	64.5	58.0	54.0
	2%	WB	49.8	48.6	50.3	54.4	60.0	63.9	66.7	66.6	63.8	60.6	55.6	52.3
		MCDB	50.2	49.5	53.2	59.8	65.3	69.2	73.1	72.1	67.1	62.3	56.5	52.9
	5%	WB	48.5	47.5	48.8	52.2	57.7	61.6	64.7	64.8	62.5	59.4	54.0	51.0
		MCDB	49.3	48.5	50.9	56.2	62.2	65.4	69.6	69.0	65.0	60.9	55.2	51.8
	10%	WB	47.2	46.3	47.6	50.6	55.8	59.9	63.2	63.5	61.2	58.3	52.2	49.3
		MCDB	48.2	47.4	49.3	53.7	59.4	63.2	67.1	66.9	63.6	59.8	53.6	50.4
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	9.3	10.5	10.6	10.3	10.4	10.4	10.1	8.7	7.7	7.0
		MCDBR	7.4	8.3	12.1	14.4	15.4	14.6	15.3	13.5	11.4	9.0	7.7	6.8
		MCWBR	7.7	7.0	8.4	8.7	9.4	9.0	8.3	7.3	7.3	6.8	7.1	7.3
	5% WB	MCDBR	7.4	7.5	10.4	12.6	13.9	13.1	13.4	11.5	9.9	7.8	7.1	6.6
MCWBR		8.2	7.0	8.0	8.3	9.3	8.9	8.2	7.3	7.2	6.9	7.3	7.4	
Clear-Sky Solar Irradiance	taub	0.328	0.344	0.380	0.398	0.406	0.406	0.404	0.396	0.376	0.361	0.335	0.320	
	taud	2.402	2.404	2.304	2.285	2.315	2.335	2.348	2.391	2.434	2.451	2.433	2.391	
	Ebn at Noon	218	247	257	266	270	270	268	265	257	238	213	201	
	Edh at Noon	20	25	33	38	38	38	37	34	29	24	19	17	
All-Sky Solar Radiation	RadAvg	255	474	843	1356	1611	1788	1706	1422	1062	603	302	196	
	RadStd	22	69	103	121	114	143	143	103	75	57	34	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page